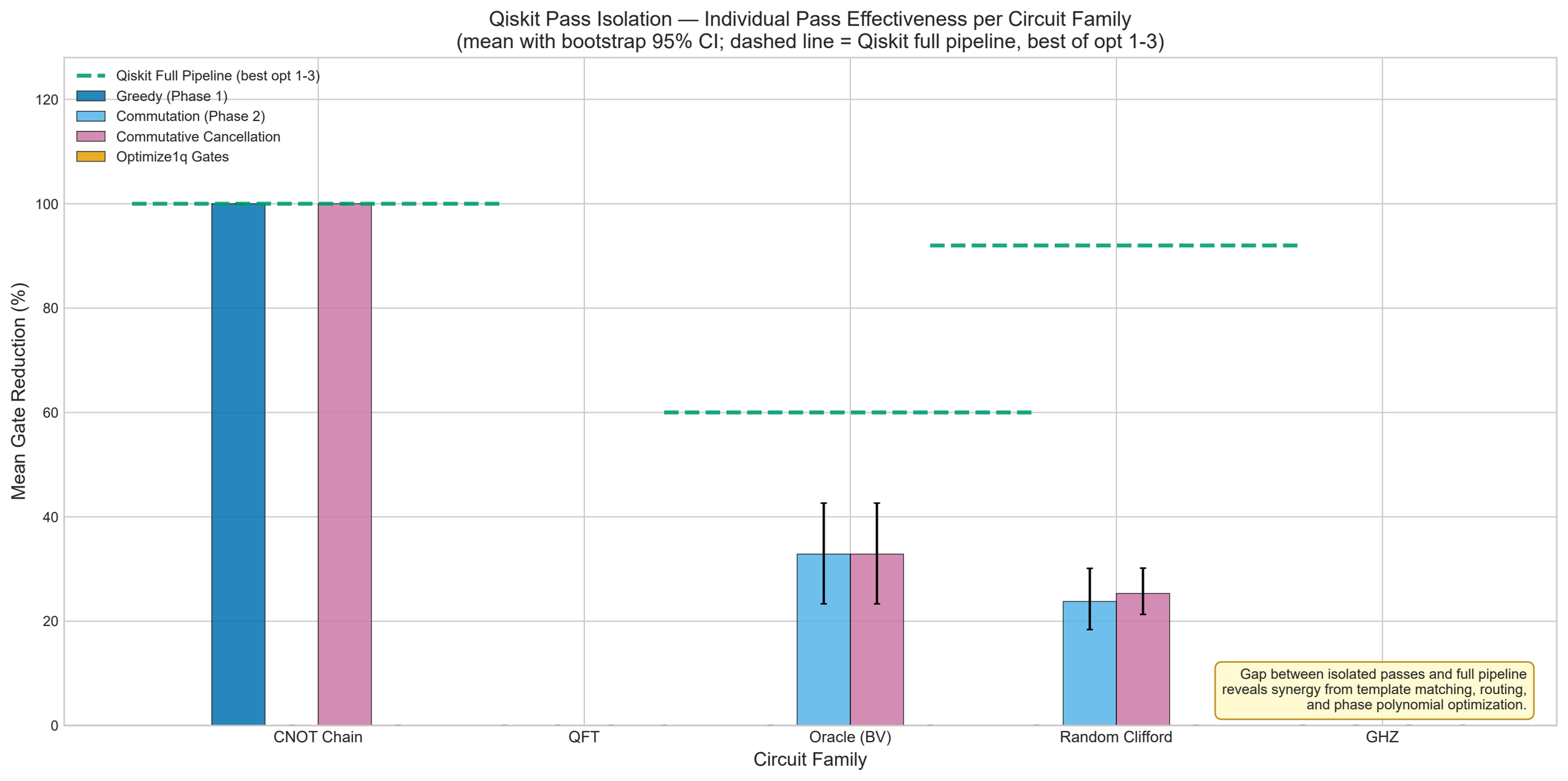
Qiskit Pass Isolation — Individual Pass Effectiveness per Circuit Family
(mean with bootstrap 95% CI; dashed line = Qiskit full pipeline, best of opt 1-3)
Qiskit Full Pipeline (best opt 1-3)
Greedy (Phase 1)
Commutation (Phase 2)
Commutative Cancellation
Optimize1q Gates
Mean Gate Reduction (%)
0
20
40
60
80
100
120
CNOT Chain
QFT
Oracle (BV)
Random Clifford
GHZ
Circuit Family
Gap between isolated passes and full pipeline reveals synergy from template matching, routing, and phase polynomial optimization.